

Dear Editor, Dear Reviewers

Thank you for considering our manuscript for publication and for providing constructive feedback.

Hereunder, you will find our detailed replies to all your comments.

The changes are highlighted in the output of the latexdiff file attached to this cover letter.

Reviewer 1

Issue 1

The contrastive learning framework needs more detailed explanation, particularly regarding how representative sequences are selected and how sequences are clustered. The description about these two parts is not very clear.

Answer

Thank you, extend the discussion to the mentioned areas

We added the following sentence to the 'Discussion'

Our pipeline increases the speed of BLAST by using a single sequence that represents a cluster, and it uses BLAST internally. The presented method is not limited to metagenomic analysis; generally, it replaces BLAST in tasks when clusters of sequences are analysed. Therefore, genome assembly, identifying cell subtypes, and phylogenetic trees might benefit from the presented approach.

Issue 2

Overall, the paper is clearly written and easy to follow for readers. They also make their code publicly available, which is good practice and will make it more widely used by the community. I do think some key details need to be expanded to justify some model choices and how well their method might generalize on new, unseen data.

Answer

Thank you for bringing this to our attention.

We selected the CNN architecture because RNN- and transformer-based methods typically require longer training times, slower inference, and greater computational resources. Our goal was to implement a relatively simple network to ensure the overall pipeline remains fast. The use of contrastive learning enabled the model to quantify the dissimilarity between sequences, which is essential for clustering them effectively. For measuring similarity, we opted for cosine dissimilarity, as it has favorable properties for separating this type of data and also slightly accelerates training <https://arxiv.org/abs/2005.10242>.

Due to the use of a specific dataset, the method has limited adaptability for generalizing to new, unseen data. Additional studies would be required to evaluate its performance on such data.

We have revised the article by modifying Section 2.1.1 with the following text:

```
\paragraph{Design Rationale}
```

```
Convolutional Neural Networks (CNNs) were chosen over Recurrent Neural  
Networks (RNNs) for faster training and inference. Contrastive learning allows  
the model to measure sequence similarity, which supports clustering.
```

and by adding Section 4.2.5, which includes the following text:

```
The model has several limitations. Its generalisation is currently restricted  
to human skin microbiome data due to the specificity of the training dataset;  
evaluation on other sample types would require additional studies. Being ANN-  
based, the method benefits from GPU acceleration, while CPU-only execution  
results in slower processing and longer pipeline runtime. This approach is  
optimised for sequences of similar length and shows limited adaptability to  
sequences of substantially different lengths. Finally, using CNNs rather than  
RNNs or transformers may limit architectural flexibility and affect  
performance, albeit providing faster training and inference times.
```

Issue 3

There are some minor grammatical errors that still remain (e.g., "This process require significant..." → "This process requires significant...", etc.)

Answer

Thank you for this issue. We rechecked the text, correcting over 30 grammatical errors and typos. The mentioned error was corrected as well.

Issue 4

Why was the Needleman-Wunsch algorithm modified as presented? Simply for speed-up? It appears to function more as an edit-distance algorithm now.

Answer

Thank you for bringing this to our attention.

The Needleman-Wunsch algorithm uses an additive scoring system, where higher values indicate greater similarity. In the contrastive learning algorithm we used, a measure of dissimilarity was required - i.e, 0 for similar sequences and higher values for dissimilar ones. Simply inverting the sign was not possible because negative values are not valid in the context of the clustering algorithm we employed. In its current form, it functions as an edit-distance algorithm.

Issue 5

How does the k-mer embedding method employed here compare to popular tools like Centrifuge, Kraken, etc.? Does it represent how this might work against state-of-the-art?

Answer

Thank you for bringing this to our attention.

In our study, the k-mer embedding method is treated as one of the comparative approaches. This method is based on the use of k-mers, a principle also employed by popular tools such as Centrifuge and Kraken. For the purposes of this study, we consider our implementation of the k-mer embedding approach to reflect current state-of-the-art practices.

Issue 6

I think the figures and tables could benefit from more descriptive titles to stand on their own. I often had to refer to the main text to fully understand them.

Answer

Thank you for this comment. We modified all figure and table captions to make them self-descriptive.

Issue 7

The term “classification quality” should be more clearly defined for readers here – especially those less familiar with the CAMI Classification Challenges. Can the authors also present performance – perhaps on an additional dataset if easier – to represent classification and speed performance? In the metagenomics use case, this could be done using standard machine learning metrics such as precision, recall, F1-score, accuracy, etc. If just focusing on clustering accuracy, alignment scores, bit scores, percent identity, etc. may be more appropriate. The use of the Jaccard index here is helpful but a bit harder to interpret, as details on the type of error (number of false positives vs. false negatives) are lost.

Answer

Thank you for bringing this to our attention.

We decided to use the Jaccard index instead of standard machine learning metrics because it is not straightforward to obtain a confusion matrix with respect to a ground truth, as the reference is the BLAST result.

Reporting false positives or false negatives would be misleading, since FPs or FNs do not make sense in this context - what appears as a false positive relative to BLAST could actually be a true positive with respect to the real ground truth. Our goal was primarily to approximate BLAST results.

We have modified the article, Section 3.5, by adding the following text:

```
as standard classification metrics relying on false positives/negatives are  
not well-defined when BLAST results serve as the reference
```

Issue 8

The authors assume in 3.3.1 that “sequence clustering is deterministic” – please elaborate here. Is this because the method actually is deterministic? Or due to the time constraints you could only perform each experimental setting once? If the latter, it would be nice to see some confidence intervals, standard error, etc. for at least a few of the mid-range sample sizes to understand how much each run might differ.

Answer

Thank you for bringing this to our attention.

In the text, we incorrectly referred to the described feature as a property of the method, whereas it is actually an implementation detail. In the experiments, a fixed seed was used. In general, the method is not deterministic, since clustering is performed by an external algorithm;

however, we do not expect significant differences within groups. Each experimental setting was run only once due to time constraints. In the case of execution time, most time is spent on the BLAST query and the clustering algorithm.

We have modified the article, Section 3.3.1 and Section 3.3.2 by adding the following text:

```
Sequences clustering is deterministic with a fixed seed.
```

Issue 9

I would like to see performance on an external validation set used with the method. The CAMI II dataset used only comes from the human skin microbiome and was further modified by the authors to create the positive and negative sequences. What specific steps were performed to prevent data leakage here? How does the method perform on a different type of sample, and are there biologically validated data that could be used to assess accuracy? If this isn't possible, it would be good for readers to know why.

Answer

Thank you for bringing this to our attention.

During the study, we ensured that the data were split into training, validation, and test sets. However, data leakage may have occurred because we did not verify the presence of duplicate records in the input data.

We have modified the article, Section 4.2.2, by adding the following text

```
Additionally, the absence of duplicate checking could have resulted in data leakage, potentially biasing the evaluation.
```

and added Section 3.4.3 with following text

```
\subsubsection{Performance on an unseen dataset}
```

```
\paragraph{Objective}
```

```
Assessing the quality of clustering on unseen data from a different sample.
```

```
\paragraph{Preparation}
```

```
We obtained an additional sample from a different body site (CAMI II Toy Human Microbiome Project\footnote{Sample 4, airways}). Processing steps were applied both to the training samples and to this new sample:
```

```

\begin{enumerate}
  \item Sequences in each sample were mapped to taxonomic identifiers
(tax\_id).
  \item From each sample, a subset of 10,000 sequences was randomly selected,
ensuring that the dominant taxon did not exceed 50\% of the subset.
  \item Sequence embeddings were generated using the ANN model.
  \item For each sample, clustering was performed with the k-medoids
algorithm, where  $k$  was chosen according to the number of true taxa in the
subset.
  \item Clusters corresponding to taxa were treated as the ground truth.
\end{enumerate}

```

The clustering performance was evaluated by computing Adjusted Mutual Information (AMI), Normalised Mutual Information (NMI), and the Fowlkes--Mallows Index (FMI) between the obtained clusters and the ground truth.

``` \paragraph{Results} ```

The evaluation metrics (AMI, NMI, FMI) for clustering performance are summarised in Table~\ref{Table:Experiment:ClusterQuality}.

``` \paragraph{Conclusions} ```

Although the clustering performance on the unseen data is not optimal, it remains comparable to the results obtained for the training data. It suggests that the model's generalisation capabilities are present but constrained, and that the outcomes are not merely random.

Issue 10

The sequences are trimmed to 150 bp – how is the trimming performed? Always from the 5' or 3' end? Randomly? What if a key motif is cleaved off – how does this impact performance? More detail is needed here on how this is performed, why 150 bp was selected, and what this means in terms of impacts on downstream classification. Further details on the k-mer size and embedding parameter choices should also be included.

Answer

Thank you for bringing this to our attention.

The description in the article is a bit redundant. In our study, we used an input dataset with reads of exactly 150 bp each, so it was not necessary to trim the sequences to this length. The

150 bp length was determined by the input dataset, but the method is applicable to sequences of any length. In the case of longer sequences, trimming should be performed from 5' to 3', if the read is shorter, it is filled by random symbols. The method is designed for sequences of similar length.

The k-mer embedding is created by constructing a vector of size 4^k , where each k-mer corresponds to a specific position in the vector. The number of occurrences of each k-mer in the sequence is then recorded at its corresponding position.

We have modified the article, Section 3.2.2, by adding the following text:

Embedding is created by constructing a vector of size 4^k , where each k-mer corresponds to a specific position in the vector. The number of occurrences of each k-mer in the sequence is then recorded at its corresponding position.

and Section 2.1.1, with following footnote:

The sequence length of 150 is a method parameter and can be adjusted depending on the application.

Issue 11

It's still unclear to me if it is better to use this method or something else like Kraken or Centrifuge for a given metagenomics classification task – it would be nice to see a direct comparison (accuracy and speed) with the final pipeline against popular methods in the field on a standardized dataset.

Answer

Kraken is fast, BLAST is more accurate but slow. Our pipeline increases the speed of BLAST by using a single sequence that represents a cluster. It uses the BLAST internally. The representative sequence allows running BLAST externally, and the searching results and their similarity metrics can be stored.

Thank you for pointing out this problem with the text. We added a text to the 'Discussion' as depicted in our answer for issue 1.

Reviewer 2

Issue 1

The contrastive learning framework needs more detailed explanation, particularly regarding how representative sequences are selected and how sequences are clustered. The description about these two parts is not very clear.

Answer

Thank you for bringing this to our attention.

After constructing a dissimilarity matrix between sequences, with non-negative values, the sequences are clustered using an external k-medoids algorithm. The medoids identified by the algorithm are considered good representatives of their respective clusters and are used in subsequent steps. In general, any algorithm that generates representatives of clusters can be used.

We have modified the article by adding Section 3.2, with the following text:

```
\subsection{Clustering Algorithm}
In the experiments, we used an external k-medoids clustering algorithm
\cite{Rust:kmedoids}. The medoids identified by this algorithm were used as
cluster representatives and used in subsequent steps.
```

Issue 2

Why do the authors use cosine dissimilarity to calculate the dissimilarity between sequence representations?

Answer

Thank you for bringing this to our attention.

We used cosine dissimilarity because it has better properties for segregating this type of data, and it also slightly speeds up training.

<https://arxiv.org/abs/2005.10242>

Issue 3

The training procedure, including hyperparameter selection, optimization algorithm, and convergence criteria, requires elaboration.

Answer

Thank you for bringing this to our attention.

The model was trained for 100 epochs using data split into training and validation sets, while a separate test set was employed in subsequent experiments. Hyperparameters were selected through grid search, resulting in the following configuration: a learning rate of 10^{-6} , weight decay of 10^{-4} , $\gamma = 0.99999$, a batch size of 256, and a dropout rate of 0.4. The optimization algorithm used was AdamW, which incorporates decoupled weight decay regularization. This choice provides more stable convergence and improved generalization compared to the standard Adam optimizer. Convergence was determined based on the absence of significant improvements in the validation loss.

We have modified the article, Section 2.1.1, by adding the following text

Learning

[...]

and the training was performed for 100 epochs. Both weight decay and the exponential learning rate schedule were employed to mitigate the risk of overfitting. The convergence criterion was defined as the absence of further improvement in the validation loss.

[...]

Parameters

[...]

The parameter search was performed using grid search.

[...]

a batch size of 256

Issue 4

The rationale for specific design choices should be justified with supporting evidence or theoretical foundations.

Answer

Thank you for bringing this to our attention. In the revised version of the paper, we have included a number of improvements and clarifications:

- We provided a better description of the training procedure
- We added a clearer explanation of hyperparameter selection and included missing hyperparameters
- We improved the descriptions of the tables for better readability

- We explained the design rationale, including why CNN was chosen, and why contrastive learning was used
- We added details about dataset sizes
- We included information about the clustering algorithm usage
- We described the process of creating k -embeddings
- We conducted an additional experiment on unseen data
- We added information about baseline method limitations
- We discussed the limitations of our model

Issue 5a

The performance evaluation metrics mainly focus on execution time and “quality”. However, the quality is not very well-defined. To evaluate metagenomic profiling, the core evaluation metrics include:

- precision: The proportion of correctly identified taxa among all predicted taxa.
- recall/sensitivity: The proportion of true taxa that were successfully identified.
- L1 Norm Error: Measure the absolute difference between predicted and true relative abundances.

Answer

Thank you for bringing this to our attention.

We decided to use the Jaccard index instead of standard machine learning metrics because it is not straightforward to obtain a confusion matrix with respect to a ground truth, as the reference is the BLAST result.

Reporting false positives or false negatives would be misleading, since FPs or FNs do not make sense in this context - what appears as a false positive relative to BLAST could actually be a true positive with respect to the real ground truth. Our goal was primarily to approximate BLAST results.

We have modified the article, Section 3.5 (p. 12), by adding the following text:

as standard classification metrics relying on false positives/negatives are not well-defined when BLAST results serve as the reference

Issue 5b

More importantly, the authors should compare their method with existing well-known methods, such as kraken2, MetaPhlAn, Centrifuge, and QIIME2. And there are several ANN-based approaches, such as VAMB, MetaDecoder, and DeepBin. Please also include these methods for performance comparison.

Answer

Thank you for pointing this out.

Our pipeline increases the speed of BLAST by using a single representative sequence for each cluster, while still relying on BLAST internally.

We compared our method with approaches based on Needleman–Wunsch alignment and k-mer embeddings. Indeed, it would also be valuable to compare with tools such as Kraken2, MetaPhlAn, Centrifuge, and QIIME2. This is planned for the next stages of the project, since such experiments require significant time (tool setup, running large-scale experiments, adapting the code, and establishing fair comparison procedures). We did not include this in the current version.

However, we believe that the experiments we have conducted already demonstrate the usefulness of our method.

Issue 6

Code Availability: The authors mention developing "ENpquizitor," a command-line tool implementing the proposed method. However, no clear information is provided about public access to the code repository. There is no reference to reproducibility materials like configuration files or trained models. I strongly recommend making the code publicly available with comprehensive documentation.

Answer

Thank you for bringing this to our attention.

We have added documentation on GitHub, including step-by-step instructions for training the model and running the experiments. Additionally, we uploaded the trained model to a cloud drive and provided the link in the README.md file."

Reviewer 3

Issue 1

The proposed method aims to convert sequences into embedding vectors. There are many existing related packages, such as DNABERT. The authors should benchmark against these tools to demonstrate that their CNN model offers an advantage.

Answer

Thank you for this valuable suggestion.

We agree that benchmarking against existing sequence embedding tools such as DNABERT would provide a stronger basis for comparison. In the current work, we focused on evaluating our CNN-based approach against classical methods. Benchmarking against DNABERT and similar models is part of our planned future work, as such experiments require considerable computational resources and careful adaptation of the pipeline.

Issue 2

There is also a problem in model training, specifically in the preparation of training data. The procedure for generating positive and negative sequences is unclear and lacks biological justification: why 0–20% mutation for positives and 20–80% for negatives, which type of DNA is used (CDS or non-coding), and whether synonymous or non-synonymous mutations are considered, since mutation rates differ among these regions. The authors need to redesign the mutation model with consideration of appropriate biological factors.

Answer

Thank you for bringing this to our attention.

The model was trained on high-quality simulated data, and the data augmentation applied was not directly biologically motivated. Although the probabilities of the generated mutations do not strictly reflect biological reality, the model was still able to perform well in our experiments. Nevertheless, the lack of incorporation of biological information may have influenced the results and potentially limited the model's overall effectiveness.

Issue 3

It is also unclear what the sizes of the training, validation, and test datasets are (Section 3.1.2).

Answer

Thank you for bringing this to our attention.

The training, validation, and test sets contain 1M, 10k, and 8192 sequences, respectively.

We have modified the article, Section 3.1.2, by adding the following text

```
The sizes of the subsets were as follows: training set - 1,000,000 sequences,  
validation set - 10,000 sequences, and test set - 8,192 sequences.
```

Issue 4

There is a problem in benchmarking. First, the baseline methods are too simple and do not consider two types of gap penalties (Gotoh algorithm) or the two types of nucleotide substitutions, transition and transversion.

Answer

Thank you for bringing this to our attention.

The baseline methods do not distinguish between different types of gap penalties, as implemented in the Gotoh algorithm, and therefore do not account for the differential treatment of insertions and deletions.

We have revised the article by adding Section 4.2.4, which includes the following text:

```
\subsubsection{Baseline Methods Limitations}  
We used simplified baseline methods, which do not account for different gap  
penalties as implemented in algorithms such as Gotoh \cite{Gotoh1982}, nor do  
they distinguish between different nucleotide substitutions, such as  
transitions and transversions.
```

Issue 5

The authors might consider using MAFFT instead of the NW algorithm, as MAFFT applies the Fourier transform and is faster.

Answer

Thank you for this hit. We plan to benchmark our tool with MAFFT. Our baseline algorithm (NW) was used mainly for quality checking. We modified the article in the section 'Future Research Directions', adding the sentence:

We plan to benchmark our method with MAFFT -- the algorithm using Fast Fourier Transform for alignment~\cite{kato2013mafft}, instead of the NW algorithm.

Issue 6

In the performance measurement, it is unclear how the reference metagenomic identification was obtained—was it determined by BLASTn?

Answer

Thank you for bringing this to our attention.

In the experiments, the reference metagenomic identification was obtained using BLAST. Although a ground truth was available, we used BLAST as the reference in order to emulate its results, since the goal of our method was to accelerate the identification process using BLAST.

We have modified the article, Section 3.5, by adding the following text

i.e., using raw BLAST results.

Issue 7

Most importantly, state-of-the-art methods are not included in the benchmark.

Answer

Thank you for pointing this out.

Our pipeline increases the speed of BLAST by using a single representative sequence for each cluster, while still relying on BLAST internally.

We compared our method with approaches based on Needleman–Wunsch alignment and k-mer embeddings. Indeed, it would also be valuable to compare with tools such as Kraken2, MetaPhlAn, Centrifuge, and QIIME2. This is planned for the next stages of the project, since such experiments require significant time (tool setup, running large-scale experiments, adapting the code, and establishing fair comparison procedures). We did not include this in the current version.

However, we believe that the experiments we have conducted already demonstrate the usefulness of our method.

Issue 8

The proposed training process is similar to the triplet neural network framework (Hoffer & Ailon, 2015, Deep metric learning using triplet network, in Similarity-Based Pattern Recognition, pp. 84–92), which classifies the data with the rule: anchor, a negative, and a positive sequence.

Answer

Thank you for bringing this to our attention.

The proposed training formula is similar to the triplet neural network framework, but it is adapted to use cosine dissimilarity.

We have added a reference to the indicated work in Section 2.1.1.

Issue 9

The authors might revise the network structure accordingly. An intuitive explanation for Equation (2) is needed, e.g., why “margin – cosine” is used for positives but “negative – margin” for negatives.

Answer

Thank you for bringing this to our attention.

Formula $[m_{pos} - s_{pos}]_+$ is used for positive pairs, because we want the loss to be zero when their similarity s_{pos} exceeds the positive margin m_{pos} . Conversely, $[s_{neg} - m_{neg}]_+$ is used for negative pairs, because we want the loss to be zero when their similarity s_{neg} is below the negative margin m_{neg} . The positive margin controls the minimum similarity between similar sequences, whereas the negative margin controls the maximum similarity between dissimilar sequences.

We have modified the article, Section 2.1.1, by adding the following text:

The positive margin controls the minimum similarity between similar sequences, whereas the negative margin controls the maximum similarity between dissimilar sequences.

Issue 10

On page 7, the value of the hyperparameter weight decay w is missing.

Answer

Thank you for pointing this out.

The weight decay is set to 10^{-4} .

We updated Section 2.1.1 accordingly.

Issue 11

Although it is good to have the GitHub link, detailed usage instructions for the package are needed.

Answer

Thank you for bringing this to our attention.

We have added documentation on GitHub, including step-by-step instructions for training the model and running the experiments. Additionally, we uploaded the trained model to a cloud drive and provided the link in the README.md file."

Reviewer 4

Issue 1

The abstract could be made more concise and sharply focused on the core contributions namely the use of contrastive learning within an ANN for representative sequence selection, its performance gains over classical methods, and the development of the Exquisitor tool.

Answer

Thank you for this comment, we updated the abstract focusing on the cored contributions. Moreover, we corrected 4 grammatical errors.

Issue 2

The manuscript does not sufficiently address the potential limitations of the proposed ANN-based system (e.g., generalizability to diverse datasets, dependence on GPU acceleration, or adaptability to varying sequence lengths). These should be discussed to present a balanced and transparent view.

Answer

Thank you for bringing this to our attention.

We have revised the article by adding Section 4.2.5, which includes the following text:

\subsubsection{Model Limitations}

The model has several limitations. Its generalisation is currently restricted to human skin microbiome data due to the specificity of the training dataset; evaluation on other sample types would require additional studies. Being ANN-based, the method benefits from GPU acceleration, while CPU-only execution results in slower processing and longer pipeline runtime. This approach is optimised for sequences of similar length and shows limited adaptability to sequences of substantially different lengths. Finally, using CNNs rather than RNNs or transformers may limit architectural flexibility and affect performance, albeit providing faster training and inference times.

Issue 3

The authors should provide more details on computational complexity, including time and memory requirements, and clarify the reliance on specific benchmark datasets during evaluation.

Answer

Thank you for bringing this to our attention.

The computational complexity of the ANN-based method depends linearly on the number of sequences, while memory usage grows quadratically with the number of sequences. Running the model requires at least 8 GB of RAM and a GPU with at least 6 GB of memory (e.g., GTX 1060) or better. It should be noted that the model was trained on a specific dataset, which limits its adaptability and generalization to new, unseen data.

We added a text to Section 4.2.5, as depicted in our answer for issue 2.

Issue 4

The literature review should be expanded to compare the proposed method with recent studies using bioinformatics datasets for classification problems, such as s13040-024-00415-8, s12859-024-05917-0, and s12859-024-05978-1. This will help situate the contribution within the current state-of-the-art.

Answer

We cited the proposed work in 'Future Research Directions' section; we now mention the feature selection techniques. We also cited the work from issue 8.

Issue 5

While the manuscript's language is generally understandable, there are grammatical and stylistic issues that should be corrected to improve readability and ensure technical clarity.

Answer

Thank you for this issue. We rechecked the text, correcting over 30 grammatical errors and typos.

Issue 6

The paper would benefit from strengthening its technical depth. Currently, some sections are presented in a more theoretical or descriptive style; these should be rewritten to include more technical detail, experimental reasoning, and methodological rigor.

Answer

Thank you for bringing this to our attention. In the revised version of the paper, we have included a number of improvements and clarifications:

- We provided a better description of the training procedure
- We added a clearer explanation of hyperparameter selection and included missing hyperparameters
- We improved the descriptions of the tables for better readability
- We explained the design rationale, including why CNN was chosen, and why contrastive learning was used
- We added details about dataset sizes
- We included information about the clustering algorithm usage
- We described the process of creating k -embeddings
- We conducted an additional experiment on unseen data
- We added information about baseline method limitations
- We discussed the limitations of our model

Issue 7

A dedicated “Model Limitations” subsection is recommended to ensure an unbiased evaluation of the work. This should highlight both the strengths and constraints of the proposed approach.

Answer

Thank you for bringing this to our attention.

We added a text to Section 4.2.5, as depicted in our answer for issue 2.

Issue 8

The “Future Work” section could be enriched by discussing how the method might be extended to large-scale bioinformatics tasks using parallel computing for big data in gene regulatory networks. For instance, parallel deep learning approaches described in s00607-025-01441-y could be considered.

Answer

Yes, we cited the proposed work in 'Future Research Directions' section, because we plan to use our tool in a distributed computing environment. We also cited one work from issue 4.

Kind regards,

Patryk Gryz, Robert Nowak

Accelerating Metagenomic Identification of DNA Sequences Using Artificial Neural Networks

Patryk Filip Gryz^{1*} and Robert Nowak^{1*}

^{1*}Institute of Computer Science, Warsaw University of Technology,
Nowowiejska 15/19, Warsaw, 00-665, Poland.

*Corresponding author(s). E-mail(s): patryk.gryz.stud@pw.edu.pl;
robert.nowak@pw.edu.pl;

Abstract

The rapid increase in the availability of DNA sequence data necessitates efficient and scalable methods for metagenomic identification. Traditional tools like BLAST, although accurate, often require significant computational resources and time, limiting their applicability to large datasets. Recent advances in artificial neural networks (ANNs) demonstrate their potential to address this issue by offering faster and more accurate solutions.

This study introduces a novel ANN-based approach for the efficient metagenomic identification of DNA sequences. The proposed method reduces the complexity of large datasets by leveraging contrastive learning within an ANN to select representative sequences. This approach enhances computational efficiency by focusing classification efforts on these representatives rather than the entire dataset. We developed Exquisitor, a command-line tool that implements the proposed method.

Comparative experiments were conducted on a benchmark dataset, evaluating the proposed ANN method alongside classical techniques such as k-mer embeddings and the Needleman-Wunsch algorithm. The presented results revealed that the ANN-based approach achieves the highest weighted average classification quality among all tested methods, outperforming both classical approaches. Additionally, the ANN method demonstrates competitive execution times, benefiting from parallelised computations enabled by GPU processing.

In conclusion, the study highlights the potential of ANN-based methods to bridge the gap between precision and scalability in genetic taxonomy.

Future research directions include enhancing the network’s flexibility for varying sequence lengths and exploring its application for dimensionality reduction in broader bioinformatics tasks.

Keywords: DNA sequencing, metagenomic identification, artificial neural networks, contrastive learning, sequence clustering

1 Background

The use of information about genetic material found in the environment, combined with sequence databases, enables metagenomic identification – the identification of organisms present in a given sample. With the development of next-generation sequencing methods[1], sequencing costs have decreased significantly, and the number of DNA sequences being ~~analyzed~~analysed has increased[2]. The time required for metagenomic identification using traditional tools, such as BLAST, is unacceptable in many applications – it can span several hours, even for relatively small collections of ~~analyzed~~analysed sequences (for example, 4096 sequences), due to the huge size and explosive growth of sequencing databases [3]. This process ~~require~~requires significant computational resources. It is necessary to create new methods to allow ~~analyzing for~~the analysis of new samples of data in an acceptable time using available resources.

The primary objective of this study is to propose ~~fast and high quality~~a fast and high-quality metagenomic identification method. We propose ~~usage~~using an artificial neural network to support the selection of representative genetic sequences. Instead of classifying ~~entire~~all environmental genetic sequences~~sample~~, the proposed approach focuses on identifying a subset of sequences that best represent the sample. These selected representatives are then used ~~to~~for identification, reducing computational complexity while maintaining accuracy.

Historically, one of the first algorithms that enabled metagenomic identification was the Needleman-Wunsch algorithm [4], developed in 1970, which allowed for the comparison of genetic sequences. However, the first solution that enabled this task within a reasonable time frame was a tool created in 1983 by D. Lipman and W. Wilbur [5]. Their method was based on dividing sequences into k -tuples, a ~~generalization~~generalisation of k -mers, and comparing them. An advancement of this approach was the BLAST algorithm, introduced in 1990 [6], which also relied on k -mers and allowed for the efficient search of similar sequences in sequence databases.

The metagenomic ~~identification~~identification became popular over the past 15 years, due to sequence database growth. A notable example of an interesting tool is the development of the MetaPhlAn tool in 2012 [7], which uses marker genes to compare the species composition of metagenomic samples. Another marker gene-based approach is mOTUs2 [8], which identifies and ~~analyzes~~analyses markers unique to specific microbial strains.

A different technique was employed in Centrifuge [9], developed in 2016, which utilizes sequence indexing based on the Burrows-Wheeler transform [10] and the Ferragina-Manzini index [11] for efficient sequences retrieval. A more classical approach is used in Kraken [12], introduced in 2014, which combines k -mers with an indexed sequence database. An unconventional method for ~~metageonomic~~metagenomic identification was proposed in 2022 with BERTax [13], which applies a transformer model

[14] to DNA sequence analysis, treating DNA as a ~~specialized~~-specialised language and allowing for classification without the need for reference databases. Another machine learning-based solution is CGRclust [15], which clusters DNA sequences using a two-dimensional chaos game representation and integrates unsupervised contrastive learning with convolutional neural networks.

Currently ~~we can propose~~, three types of methods have been developed for metagenomic identification: methods using alignment, methods based on k -mers and methods using marker genes. Alignment-based tools should yield the best results, but they require significant computational resources. Algorithms based on k -mers are very fast, but they do not take into account all the information about the structure of the input genetic sequences, which reduces their quality. Solutions using marker genes are ~~characterized~~-characterised by good quality, but they allow analysis only for subsets of organisms for which marker genes are known.

Although recent years have seen studies on metagenomic identification using machine learning methods, there is still a lack of research focusing on the application of artificial neural networks with contrastive learning for efficient sequence clustering. In many fields, artificial neural networks (ANN) have demonstrated the ability to outperform classical heuristic algorithms, offering faster and more accurate results. By leveraging the capabilities of ANN, this ~~works~~-work aims to accelerate the metagenomic identification process while maintaining at least the same level of quality as traditional methods. Such an approach could significantly accelerate the classification process using available tools.

The rest of the paper is ~~organized~~-organised as follows. Section 2 defines the problem, presents the proposed approach, and provides implementation details of the developed tool Exquisitor. Section 3 describes the experimental setup, including ~~used datasets~~-the datasets used, the performance metrics for evaluation, and the results obtained from experiments with analysis. Finally, section 4 concludes the paper by ~~summarizing~~-summarising the main findings, ~~discuss on the limitations of~~-discussing limitations of the proposed approach and suggesting potential directions for future research.

2 Methods

The core component of the proposed tool, Exquisitor, is a processing pipeline designed with flexibility in mind, allowing the integration of various DNA sequence clustering methods. The pipeline consists of a series of steps executed sequentially to perform metagenomic identification. A schematic representation of the pipeline is shown in Figure 1, where input and output data are represented by circles, and ~~each processing step is represented by a rectangle~~-a rectangle represents processing step. The pipeline is composed of the following five steps:

1. Preparation of DNA Sequences

This step is responsible for verifying the correctness of the provided DNA sequences and aligning them to a required length, which may be necessary for subsequent processing.

2. Dissimilarity Calculation

~~In this step, the dissimilarity between DNA sequences is calculated~~ using one of three available algorithms: two classical methods and one custom-developed approach. The resulting dissimilarity matrix serves as the basis for clustering.

3. Sequence Clustering

Based on the previously computed dissimilarities, sequences are grouped into clusters. Each cluster consists of one representative sequence and several member sequences. The main objective of this step is to reduce the number of sequences processed in later stages by selecting the most informative representatives.

4. Database Search

In this step, the representative sequences are queried against a reference database to identify similar sequences. This allows the system to infer the likely origin of the sequences and determine the organisms they may belong to.

5. Postprocessing

~~In this final step, information from the clusters is integrated~~ with the database search results.

2.1 Exquisitor: Method Description

The proposed method ~~involves reducing~~ uses an artificial neural network to reduce the dimensionality of input DNA sequences to a feature vector in the form of $\mathbb{R}^{64}$ ~~using an artificial neural network~~. The neural network employs contrastive learning, ~~which enables the learning of representations~~ enabling representations to be learned while preserving dissimilarity between sequences. The dissimilarity between sequence representations will be calculated using cosine dissimilarity, expressed by the formula in equation (1).

$$dsim(A, B) = 1 - sim(A, B) \quad (1)$$

where:

$$sim(A, B) = \cos \theta = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \cdot \sqrt{\sum_{i=1}^n B_i^2}}$$

A, B —compared vectors,

θ —angle between A and B ,

A_j, B_j — j -th element of the vector A and B , respectively.

2.1.1 Artificial Neural Network

Architecture

The ANN model consists of two parts. The first part comprises two convolutional blocks with batch ~~normalization~~ normalisation and is responsible for feature extraction from the sequence. The second part ~~utilizes~~ utilises fully connected layers with dropout and the GELU activation function [16] and is connected to the first part of the model via a flattening layer.

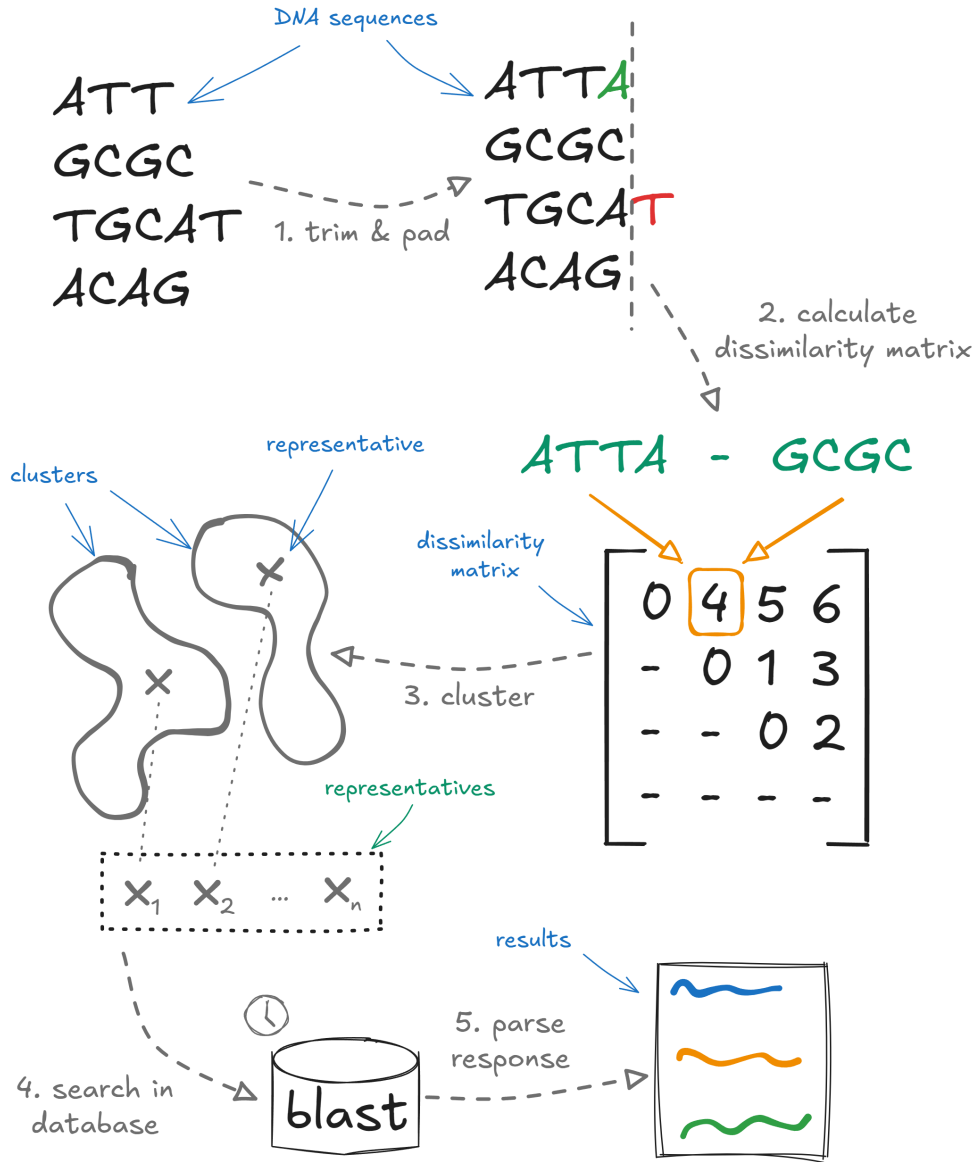


Fig. 1 Diagram Overview of the pipeline: preparation of DNA sequences, dissimilarity calculation, sequence clustering, database search and postprocessing.

The model output is the output of the final linear layer, which is a feature vector of dimension $\mathbb{R}^{64}$.

Schematically, the architecture is presented in Figure 2.

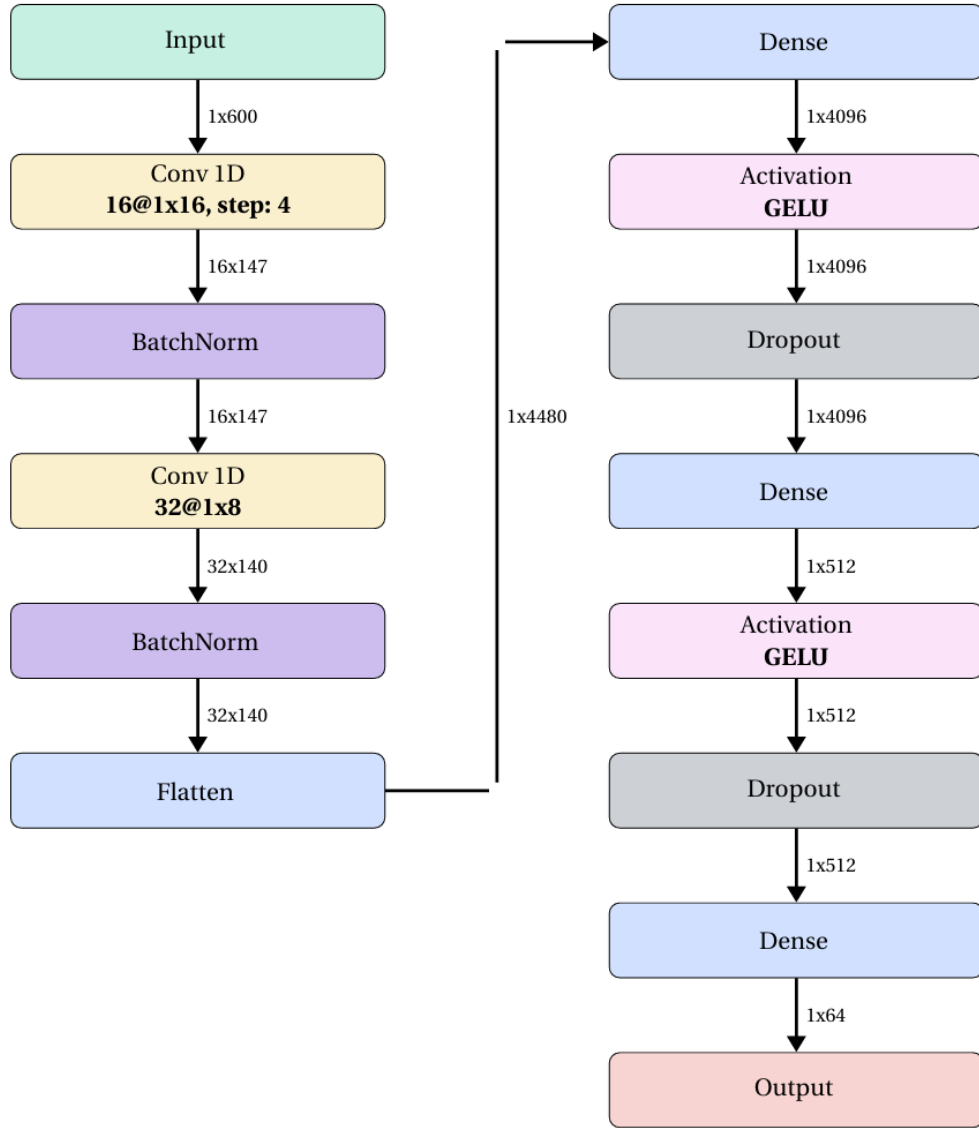


Fig. 2 Diagram of ANN architecture. Two parts were depicted: the convolutional blocks (on the left) are responsible for feature extraction, and fully connected layers with dropout to generate the results.

Design Rationale

Convolutional Neural Networks (CNNs) were chosen over Recurrent Neural Networks (RNNs) for faster training and inference. Contrastive learning allows the model to measure sequence similarity, which supports clustering.

Input

The input to the model consists of DNA sequences of length 150¹, which are encoded into a vector of dimensions 1×600 using one-hot encoding [17].

Training Examples

The training and validation examples consist of an anchor, a positive sequence ~~that is~~ similar to the anchor, and a negative sequence ~~that is~~ dissimilar to the anchor.

Dataset

The training and validation datasets were created based on the first sample of genetic sequences from the CAMI II Toy Human Microbiome Project dataset [18]. The sample contains simulated metagenomic data from the human skin microbiome. Examples were obtained by randomly selecting anchors from the dataset and modifying these anchors to create positive and negative sequences. The modification involved a point mutation of a given nucleotide to another one. In the case of positive sequences, the change affected between 0% and 20% of the anchor length, whereas for negative sequences, the change ranged from 20% to 80% of the anchor length.

Loss function

The loss function used is defined as:

$$\text{Contrastive loss} = [m_{pos} - s_{pos}]_+ + [s_{neg} - m_{neg}]_+ \quad (2)$$

where:

m_{pos} – similarity margin between the positive sequence and the anchor,

m_{neg} – similarity margin between the negative sequence and the anchor,²

s_{pos} – cosine similarity of the positive sequence to the anchor,

s_{neg} – cosine similarity of the negative sequence to the anchor.

The proposed loss function is similar to the triplet neural network framework[19], but it is adapted to use cosine dissimilarity.

Learning

The model was trained on a dataset of 10^6 training examples and 10^4 validation examples. The ~~optimizer~~ optimiser used was *AdamW* [20], and the training was performed for 100 epochs. Both weight decay and the exponential learning rate schedule were employed to mitigate the risk of overfitting. The convergence criterion was defined as the absence of further improvement in the validation loss.

Quality

As a measure of model quality, the contrastive loss of the model was computed on the validation set.

¹The sequence length of 150 is a method parameter and can be adjusted depending on the application.

Parameters

Experiments were conducted to determine the optimal training parameters for the artificial neural network model. The tested parameters included the learning rate coefficient λ , weight decay w , the coefficient γ used in the exponential learning rate decay, the dropout rate, and the effectiveness of using three perceptron layers. [The parameter search was performed using grid search.](#)

As a result of these experiments, the best parameters were selected:

$$\lambda = 10^{-6}, \quad w = 10^{-4}, \quad \gamma = 0.99999$$

a dropout rate of 0.5, [a batch size of 256](#), and a confirmed positive impact of using three perceptron layers.

Additionally, the loss function was used with the following parameters:

$$m_{pos} = 1.0, \quad m_{neg} = 0.25.$$

Results of Learning

As a result of training the neural network, the model achieved a validation set loss of 0.17.

2.2 Exquisitor: Implementation Details

The solution consists of two components. The first is a library that contains components enabling the creation of pipelines for genetic material analysis in the form of metagenomic identification. The second component is a console application, which uses the library's components to ~~allow for the creation and configuration of~~ [create and configure](#) processing pipelines. ~~Additionally, a~~ [A](#) web application was ~~created, which allows also created, allowing~~ users to submit analysis tasks through a web browser.

The console application was implemented using the *clap* library[21]. In [the](#) library the *k*-medoid algorithm was provided by the *kmedoids* library[22], and the neural network was built using the *burn* library[23] with the *wgpu* execution environment. Web application was implemented using the *axum* library[24] with *tokio*[24] async environment.

Additionally, the following tools were used in the work:

- *cargo* as the package manager and build system in Rust;
- *rustup* for automatic management of Rust versions;
- *clippy* for static code analysis in Rust;
- *rustfmt* for automatic formatting of Rust source code;
- *cargo test* for conducting unit tests;
- *git* as the version control system, enabling change tracking and code history management.

3 Results

This ~~sections~~ section presents the results of the experiments conducted to evaluate the proposed solution (Exquisitor), which aims to improve performance while maintaining the quality of metagenomic ~~identifaeation~~ identification. The study ~~involved~~ ~~comparing~~ compared the new approach with two classical methods using the same dataset. The evaluation focused on execution time and identification quality to assess the effectiveness of the proposed approach.

3.1 Datasets

3.1.1 Dataset Description

In the experiments, the *CAMI II Toy Human Microbiome Project* [18] dataset was used, which is the same dataset used for training the artificial neural network model. ~~Dataset was chosen~~; The dataset was chosen because it was created for benchmarking bioinformatics tools and contains a large number of sequences, enabling its use both in experiments and in the learning process.

Subsets of sequences were created from the dataset with sizes expressed by the formula 2^k for $k \in [0, 12]$. Subsets are disjoint, and only those sequences ~~that were~~ not used for training the ANN model were employed in their construction.

3.1.2 Dataset Preparation

The subsets were created by randomly sampling, without replacement, the indices of genetic sequences from the reference dataset that were to be included in each subset. Indices of sequences used in the ANN training and validation sets ~~of the ANN~~ were excluded from the sampling. The sizes of the subsets were as follows: training set - 1,000,000 sequences, validation set - 10,000 sequences, and test set - 8,192 sequences.

3.2 Clustering Algorithm

In the experiments, we used an external k-medoids clustering algorithm [22]. The medoids identified by this algorithm were used as cluster representatives and used in subsequent steps.

3.3 Base-line Algorithms

Two classical approaches were implemented and used ~~for comparison with the~~ to compare their results with the results of the proposed neural network.

3.3.1 Modified Needleman-Wunsch Algorithm

The first classical approach is the Needleman-Wunsch algorithm. This algorithm allows for global alignment of genetic sequences. It uses a similarity matrix between sequences, which is constructed according to the rules given in equation (3). A modification of the approach presented in equation (4) is applied in the work.

$$\begin{aligned}
D_{i,0} &= i \cdot g, & \text{for } i \in [1, n+1] \\
D_{0,j} &= j \cdot g, & \text{for } j \in [2, m+1] \\
D_{i,j} &= \max \begin{cases} D_{i-1,j} + g \\ D_{i,j-1} + g \\ D_{i-1,j-1} + s(A_i, B_j) \end{cases}, & \text{for } i \in (1, n+1] \text{ and } j \in (1, m+1]
\end{aligned} \tag{3}$$

where:

A, B —compared sequences,
 n, m —lengths of sequences A and B ,
 D —similarity matrix of size $n+1 \times m+1$,
 $g \in \mathbb{R}$ —penalty for a gap,
 $s(A_i, B_j) \in \mathbb{R}$ —similarity between the i -th element in sequence A
and the j -th element in sequence B .

$$D_{i,j} = \min \begin{cases} D_{i-1,j} + g \\ D_{i,j-1} + g \\ D_{i-1,j-1} + s(A_{i-1}, B_{j-1}) \end{cases}, \quad \text{for } i \in (1, n+1] \text{ and } j \in (1, m+1] \tag{4}$$

In the solution, the following parameter values were adopted:

$$\begin{aligned}
g &= 2, \\
s(a, b) &= \begin{cases} 0, & \text{for } a = b, \\ 1, & \text{for } a \neq b. \end{cases}
\end{aligned}$$

3.3.2 k -mer Embeddings

The second approach ~~will be the use of~~ use k -mers as embeddings, which allows for the representation of DNA sequences as numerical vectors. Embedding is created by constructing a vector of size 4^k , where each k -mer corresponds to a specific position in the vector. The number of occurrences of each k -mer in the sequence is then recorded at its corresponding position. These vectors will then be compared using Euclidean distance.

Table 1 Metagenomic identification execution time [comparison of presented solution \(NP\) and benchmark methods \(NW, \$k\$ -mer, ANN\)](#).

Number of sequences	Execution time [s]			
	NP	NW	k -mer	ANN
1	7107	6025	6087	6077
2	6129	5994	5988	6035
4	6108	5983	6024	6019
8	6196	5988	5925	6014
16	6290	5941	5946	6118
32	6279	6035	5962	6092
64	6316	6113	6107	6139
128	6560	6238	6206	6233
256	6753	6446	6295	6363
512	7091	6972	6326	6347
1024	8059	8743	6248	6269
2048	8517	16461	6472	6332
4096	9433	timeout	7003	6550

3.4 Experiments

3.4.1 Execution Time of Metagenomic Identification with Exquisitor

Objective

Measuring the execution time of metagenomic identification using different genetic sequence clustering methods.

Assumptions

1. Sequences clustering is deterministic [with a fixed seed](#).
2. Metagenomic identification is the only task running on the machine.

Results

As a result of the experiment, the execution time for metagenomic identification was obtained for all methods. In the case of the method using the modified Needleman-Wunsch algorithm, results for metagenomic identification of 4096 sequences could not be obtained due to exceeding the execution time. Figure 3 shows a graph of the metagenomic identification execution time as a function of the number of input sequences for: the method with the modified Needleman-Wunsch algorithm (NW), the method using k -mer embeddings (k -mer), the method using artificial neural networks (ANN), and for metagenomic identification of all sequences without using a pipeline (NP). Detailed execution times are provided in Table 1.

Conclusions

The method using the modified Needleman-Wunsch algorithm exhibited the fastest increase in execution time for metagenomic identification. The sharp rise is due to the time-consuming process of comparing sequences with each other. Despite reducing the

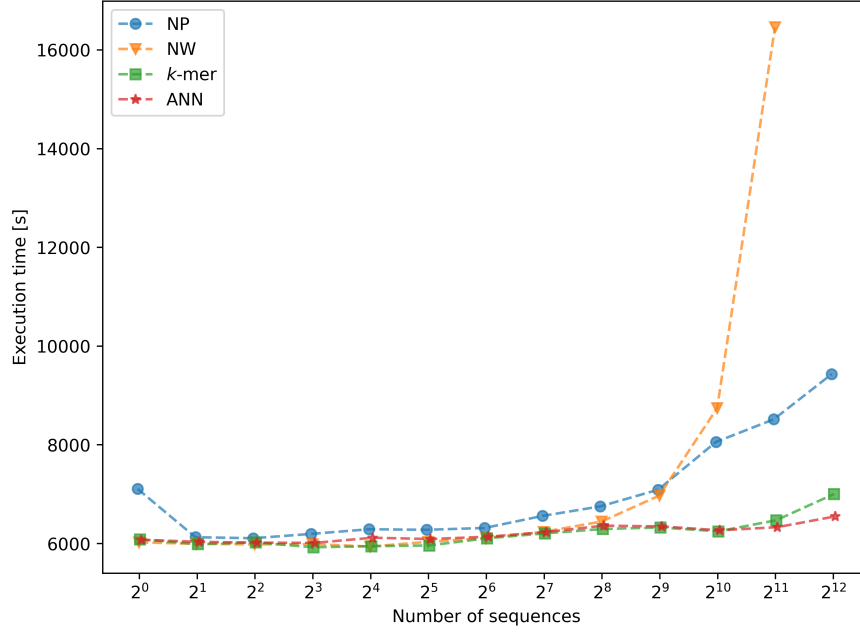


Fig. 3 ~~Metagenomic~~The comparison of the metagenomic identification execution time by the presented solution (NP) and benchmark methods: modified Needleman-Wunsch algorithm (NW), k -mer embeddings (k -mer), and a method using an artificial neural network (ANN).

number of ~~sequences being classified~~classified sequences, the execution time using this method exceeded that of metagenomic identification for all sequences.

The methods using k -mer embeddings and ANN showed comparable execution times, with a slight advantage for the former. Both methods use sequence representations for comparison, resulting in a slower increase in the time needed to build the dissimilarity matrix. These methods performed faster than metagenomic identification for all sequences. In the case of the ANN, the shorter execution time for 4096 sequences could be due to the simultaneous computation of embeddings for all sequences using the GPU.

3.4.2 Quality of Metagenomic Identification with Exquisitor

Objective

Examining the quality of metagenomic identification using the implemented methods in comparison to the metagenomic identification of all sequences.

Assumptions

1. Sequences clustering is deterministic with a fixed seed.
2. BLASTn is deterministic and always returns the same results for a given sequence and specified parameters.
3. In the case where it is not possible to compute the quality for a given metagenomic identification run, a quality value of 0 is assumed. This value is not considered when calculating the weighted average quality.

3.5 ~~Exquisitor Performance Measures~~

Exquisitor Performance Measures

Quality of metagenomic identification was measured using a modified Jaccard index, expressed by equation 5, as standard classification metrics relying on false positives/negatives are not well-defined when BLAST results serve as the reference.. The measure was used to compare the quality of metagenomic identification performed using the implemented methods against metagenomic identification without ~~the use of a processing pipeline~~ using a processing pipeline, i.e., using raw BLAST results.

$$Q = \frac{\sum_{r \in (O(R) \cap O(E))} R_r}{\sum_{r \in (O(R) \cup O(E))} R_r} \quad (5)$$

where:

R – set of reference results,

E – set of obtained results,

R_r – number of results in reference set assigned to the organism r

$O(X)$ – set of unique organisms, for which results were assigned in the set X

~~To compare the quality of multiple metagenomic identification runs, the~~ The weighted average quality, defined by equation 6, was used to compare the quality of multiple metagenomic identification runs.

$$Q_{\text{avg}} = \sum_{c \in C} \frac{n_c}{n} Q_c \quad (6)$$

where:

C – set of metagenomic identification runs,

Q_c – quality of metagenomic identificationn c ,

n_c – number of input sequences for metagenomic identification c ,

n – number of input sequences $n = \sum_{c \in C} n_c$.

Results

The ~~quality of the classification~~ classification quality was calculated using the measure expressed by equation 5 for the implemented methods ~~in comparison~~ compared to the

full metagenomic identification. Figure 4 shows a graph of classification quality as a function of the number of input sequences, with detailed quality results provided in Table 2. The weighted average quality calculated using equation 6 for the method with the modified Needleman-Wunsch algorithm was 0.899, for the k -mer embedding method it was 0.921, and for the method using the artificial neural network, it was 0.944.

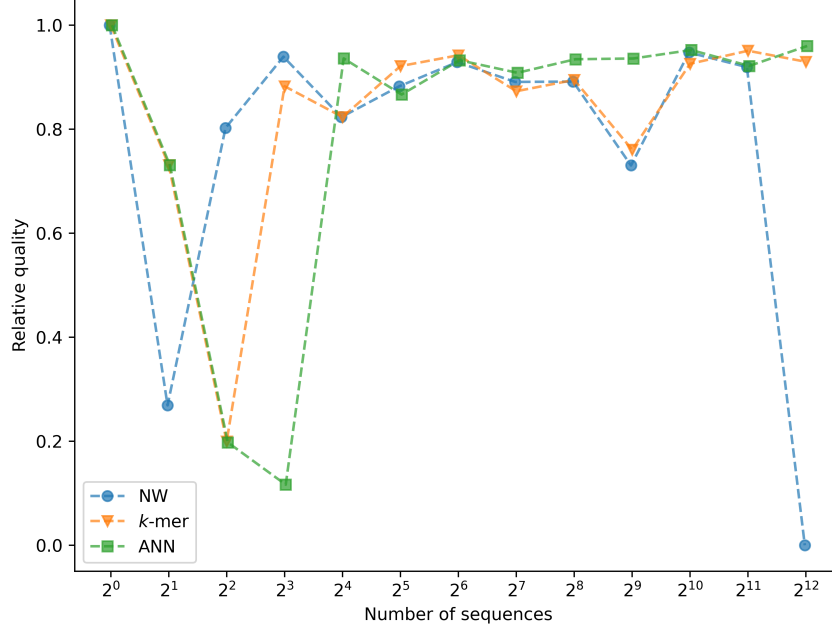


Fig. 4 Metagenomic-Comparison of the metagenomic identification quality returned by the presented method with the benchmark methods.

Conclusions

All implemented methods achieved very good classification quality results, exceeding 0.7 in cases where the number of groups was significantly larger than the number of input sequences. The weighted average quality for all methods also reached a high value.

The method using the artificial neural network achieved the best weighted average quality, slightly outperforming the k -mer embedding method and the method with the modified Needleman-Wunsch algorithm. The artificial neural network method owes its

Table 2 ~~Metagenomic identification~~ Comparison of the metagenomic identification quality returned by the presented method with the benchmark methods.

height	Number of sequences	Method		
		NW	k -mer	ANN
	1	1.0	1.0	1.0
	2	0.27	0.73	0.73
	4	0.8	0.2	0.2
	8	0.94	0.88	0.12
	16	0.82	0.82	0.94
	32	0.88	0.92	0.87
	64	0.93	0.94	0.93
	128	0.89	0.87	0.91
	256	0.89	0.89	0.93
	512	0.73	0.76	0.94
	1024	0.95	0.93	0.95
	2048	0.92	0.95	0.92
	4096	0.0	0.93	0.96

result to the model’s consideration of the full sequence structure, which allowed for a better determination of dissimilarity between sequences.

3.4.1 Performance on an unseen dataset

Objective

Assessing the quality of clustering on unseen data from a different sample.

Preparation

We obtained an additional sample from a different body site (CAMI II Toy Human Microbiome Project³). Processing steps were applied both to the training samples and to this new sample:

1. Sequences in each sample were mapped to taxonomic identifiers (tax_id).
2. From each sample, a subset of 10,000 sequences was randomly selected, ensuring that the dominant taxon did not exceed 50% of the subset.
3. Sequence embeddings were generated using the ANN model.
4. For each sample, clustering was performed with the k-medoids algorithm, where k was chosen according to the number of true taxa in the subset.
5. Clusters corresponding to taxa were treated as the ground truth.

The clustering performance was evaluated by computing Adjusted Mutual Information (AMI), Normalised Mutual Information (NMI), and the Fowlkes–Mallows Index (FMI) between the obtained clusters and the ground truth.

Results

³Sample 4, airways

Table 3 Comparison of clustering quality on seen and unseen data.

Sample	AMI	NMI	NMI
Sample 1 (seen)	0.288	0.286	0.424
Sample 4 (unseen)	0.219	0.243	0.379

The evaluation metrics (AMI, NMI, FMI) for clustering performance are summarised in Table 3.

Conclusions

Although the clustering performance on the unseen data is not optimal, it remains comparable to the results obtained for the training data. It suggests that the model's generalisation capabilities are present but constrained, and that the outcomes are not merely random.

4 Discussion

4.1 Interpretation of Results

The results of the experiments partially confirmed the expectations set at the beginning of the research. All methods achieved satisfactory metagenomic identification quality. The method ~~utilizing~~ ~~utilising~~ the artificial neural network performed particularly well, achieving the best weighted average classification quality while maintaining execution time comparable to the method based on k -mer embeddings. The low metagenomic identification quality with a small number of input sequences may be due to the inappropriate selection of group representatives, caused by too few available sequences. Therefore, the implemented methods should only be used in cases where the number of input sequences exceeds a certain threshold, and their use significantly reduces the time required for the metagenomic identification process. For the data tested, this condition is met for 128 sequences, where the metagenomic identification quality was no less than 0.87, and the use of any grouping method reduced the time by at least 5 minutes, resulting in a 5% acceleration compared to the metagenomic identification of all sequences. The use of the artificial neural network allowed for an improvement in classification quality compared to other methods. In addition to achieving good classification quality, it does not lag behind classical methods.

Our pipeline increases the speed of BLAST by using a single sequence that represents a cluster, and it uses BLAST internally. The presented method is not limited to metagenomic analysis; generally, it replaces BLAST in tasks when clusters of sequences are analysed. Therefore, genome assembly, identifying cell subtypes, and phylogenetic trees might benefit from the presented approach.

4.2 Limitations

4.2.1 Experiments Duration

The full ~~process of~~ metagenomic identification of DNA sequences for each method and each experimental subset took approximately 5 days, which led to the limitation of the number of experiments to a single run and subset size to 4096 sequences.

4.2.2 Single Dataset

In work, only one sample from the chosen dataset was used. The selection of a sample containing DNA sequences from the human skin microbiome might have limited the space of ~~analyzed~~ analysed sequences, which could have influenced the experimental results, as ~~the~~ both the artificial neural network and the experiments used DNA sequences from the same microbiome. Additionally, the absence of duplicate checking could have resulted in data leakage, potentially biasing the evaluation.

4.2.3 Single Dissimilarity Matrix

~~The use of~~ Using a single dissimilarity matrix for all sequences limits the number of input sequences, as the matrix build time ~~of matrix~~ scales quadratically with the number of sequences.

4.2.4 Baseline Methods Limitations

We used simplified baseline methods, which do not account for different gap penalties as implemented in algorithms such as Gotoh [25], nor do they distinguish between different nucleotide substitutions, such as transitions and transversions.

4.2.5 Model Limitations

The model has several limitations. Its generalisation is currently restricted to human skin microbiome data due to the specificity of the training dataset; evaluation on other sample types would require additional studies. Being ANN-based, the method benefits from GPU acceleration, while CPU-only execution results in slower processing and longer pipeline runtime. This approach is optimised for sequences of similar length and shows limited adaptability to sequences of substantially different lengths. Finally, using CNNs rather than RNNs or transformers may limit architectural flexibility and affect performance, albeit providing faster training and inference times.

4.3 Possible Improvements

4.3.1 Architecture of Artificial Neural Network

The proposed artificial neural network architecture is not flexible and is only suitable for ~~analyzing~~ analysing sequences of similar lengths. It would be possible to modify the model's architecture to incorporate recurrent neural networks (RNNs) and transformers, enabling ~~the processing of~~ processing sequences with varying lengths. RNNs and transformers could replace the first part of the model, which currently relies on

convolutional layers. They would be used to create embeddings, ~~which would later be processed by linear layers~~ linear layers would later process to generate an embedding that preserves dissimilarity properties.

4.3.2 Batching Sequences

Creating a single dissimilarity matrix for very large samples is not optimal, as it significantly impacts the ~~performance of the method~~ method's performance. To address this issue, pre-clustering the sequences into batches can provide a solution. Processing in batches may reduce quality, but allows for maintaining a short execution time.

4.3.3 ~~Automated~~ Automated Determination Of The Number of Groups

~~Current~~ Our approach takes the number of created groups as a parameter, which can influence the quality of the results. For very similar sequences, the results are not improved when a large number of groups is ~~excessive and does not improve the results~~ set. Conversely, ~~for highly diverse sequences~~, a low number of groups can reduce quality for highly diverse sequences. Automating the determination of the optimal number of groups could solve the issue and minimize the need for fine-tuning.

5 Conclusions

5.1 Summary of Findings

The experiments demonstrated that the proposed method based on artificial neural networks outperforms traditional approaches in terms of quality, while maintaining execution times comparable to those of the classical k -mer method. ~~This~~ It highlights the potential of neural networks for improving performance without significant computational overhead.

5.2 Future Research Directions

Future research could focus on ~~utilizing~~ utilising artificial neural networks as an independent tool for metagenomic identification. Another promising direction is the application of artificial neural networks for dimensionality reduction of genetic sequences, which may improve efficiency and performance in various analyses. We also plan to apply the feature selection techniques successfully in other bioinformatics pipelines [26]. Planned development efforts are focused on providing interfaces for the Exquisitor tool, enabling its integration with existing popular metagenomic pipelines. Moreover, distributed computing and cluster computing platforms will be provided [27]. We plan to benchmark our method with MAFFT – the algorithm using Fast Fourier Transform for alignment [28], instead of the NW algorithm.

6 Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and material

Exquisitor is distributed under the terms of ~~both~~ the MIT license and the Apache License (Version 2.0). The source code is available at <https://github.com/pfgryz/exquisitor>.

Competing interests

Authors ~~declare~~ declare no competing interests.

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Authors' contributions

PG and RN identified the problem, PG designed the approach, downloaded the data, implemented the software, performed numerical experiments, PG and RN prepared the draft. All authors have read and agreed to the published version of the manuscript.

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Not applicable.

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